Maximising Edge Density in Graphs Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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Maximising edge density in graphs under connectivity constraints

↓ *translation*

networks with many links and low connectivity

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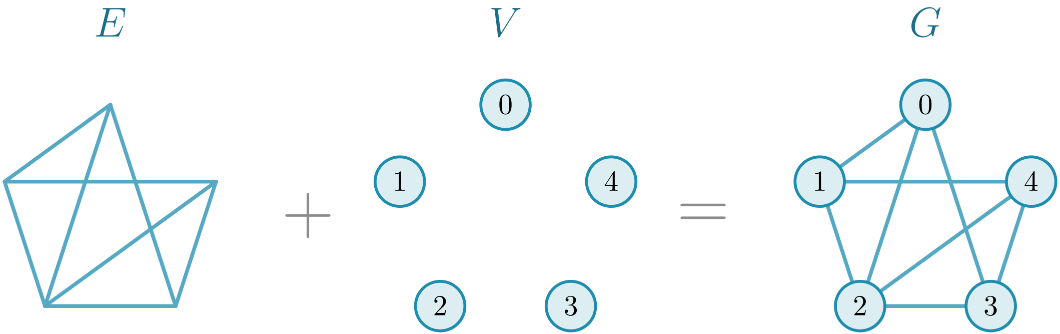
Graph

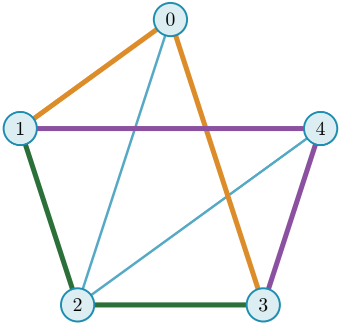


Erdős Problem 915

$$\ell_m(n) \;=\; \max_{|V(G)|=n}\Big\{\,|E(G)| \;:\; \boxed{\lambda_G^{\max}} \le\; m-1\Big\}$$

$$\lambda_G^{\max} \;=\; \max_{u\neq v} \boxed{\lambda_G(u,v)}$$

$\lambda_G(u,v) \;=$ the largest number of u–v routes that share no edge: one colour each

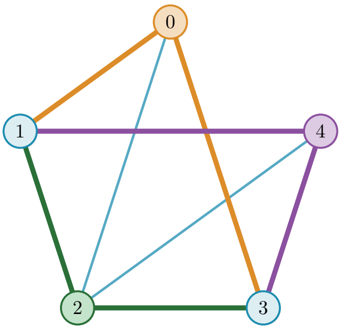


The same question, vertex by vertex

$$k_m(n) \;=\; \max_{|V(G)|=n}\Big\{\, |E(G)| \;:\; \boxed{\kappa_G^{\max}} \le\; m-1 \,\Big\}$$

$$\kappa_G^{\max} \;=\; \max_{u\neq v} \boxed{\kappa_G(u,v)}$$

$\kappa_G(u,v) \;=\;$ the largest number of u–v routes that share no interior vertex



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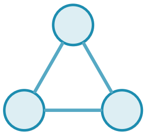
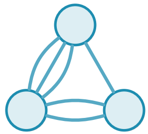
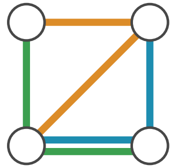


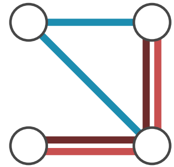
simple graph



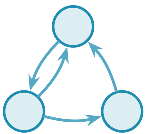
multigraph



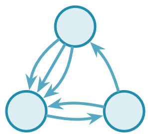
hypergraph



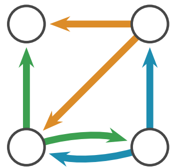
multihypergraph



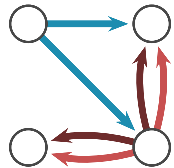
simple digraph



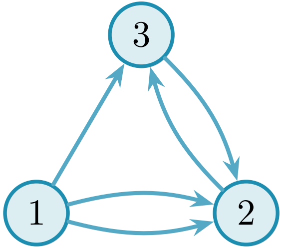
directed multigraph



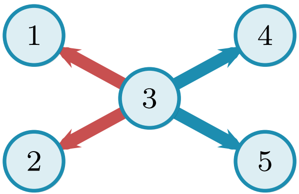
directed hypergraph



directed multihypergraph



$$\longrightarrow \begin{pmatrix} \color{red}{0} & 2 & 1 \\ 0 & \color{red}{0} & 1 \\ 0 & 1 & \color{red}{0} \end{pmatrix}$$



$\longrightarrow$

list of hyperedges:

$e_1 : \{3 \mid 1, 2\}$

$e_2 : \{3 \mid 4, 5\}$

$$\begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

INPUT

||

directed multigraph

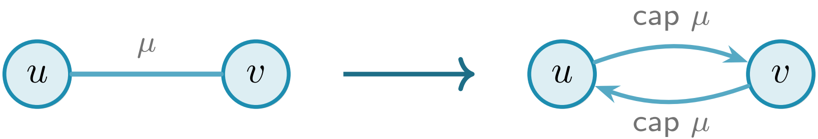
max flow

over every pair

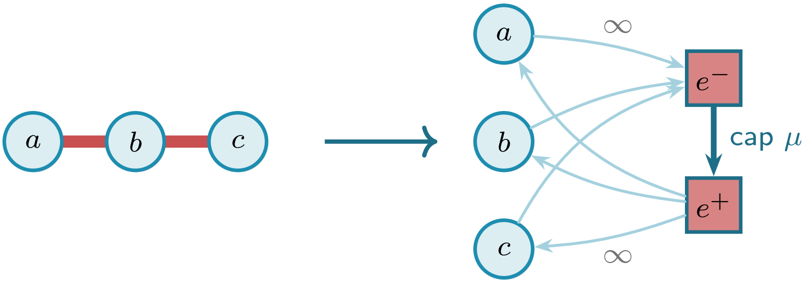
$$\lambda_G^{\max} = 3$$

OUTPUT

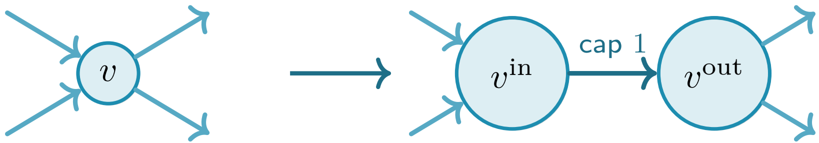
undirected $\rightarrow$ directed



hypergraph $\rightarrow$ non-hypergraph



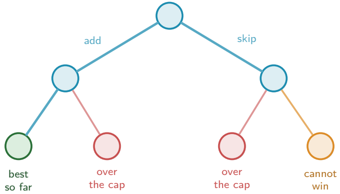
$$\boldsymbol{\kappa} \;\longrightarrow\; \boldsymbol{\lambda}$$



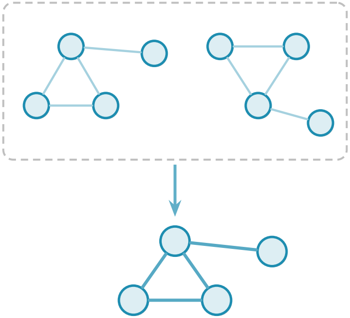
every intermediate vertex is split this way
the flow itself runs from s^{out} to t^{in}

Efficiency: three ways to make the sweep finish

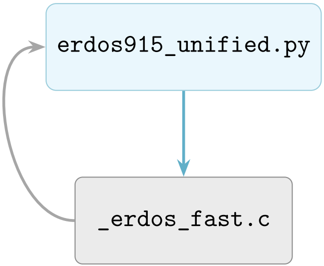
pruning



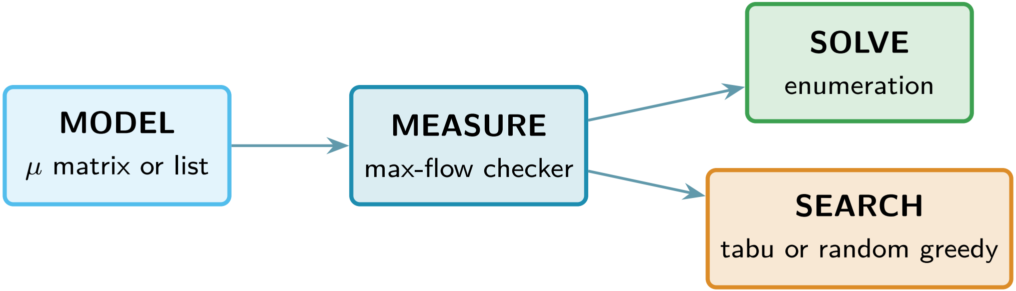
generating search



C: faster max flow



Solving versus searching

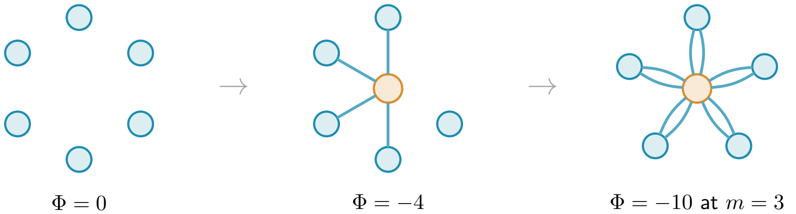


The search: one number per graph

$$\Phi(G) \;=\; \underbrace{-\,|E(G)|}_{\text{reward}} \;+\; \underbrace{\Lambda \cdot \max\Big(0,\; c^{\max}(G) - (m-1)\Big)}_{\text{penalty}}$$

reward
more edges, lower energy

penalty
discourages excess connectivity



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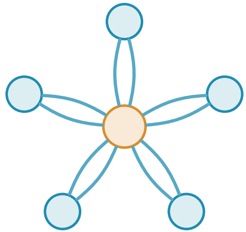
Extremal families the search rediscovers

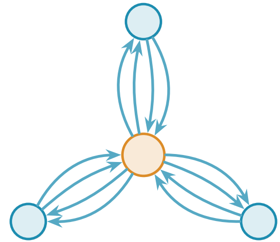
tree

$n = 6,\ \ell_2(n) = 5$



multigraph star

$n = 6,\ m = 3,\ L_3(n) = 10$

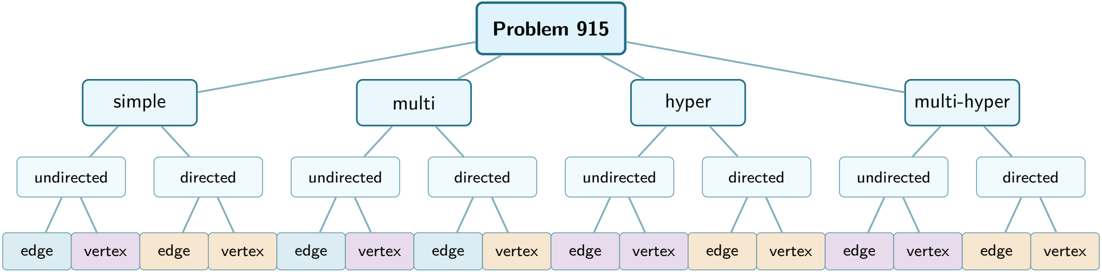


double star

$n = 4,\ m = 3,\ L_3^{\mathrm{dir}}(n) = 12$

examples of the families, not the saved search witnesses

Status of the sixteen variants



exact value established in the graph domain

general exact value open, bounds and small cases known

partial results or a conjectural general formula

Thank you

github.com/louisvandenbruwaene/thesis

Backup: the sixteen variants

HOW TO READ THE NEXT SIXTEEN FRAMES

Each frame takes one variant, fixes a small size, and evaluates the formula there. Top right it carries its leaf's colour from the status tree.

EXACT VALUE ESTABLISHED PARTIAL OR CONJECTURAL GENERAL VALUE OPEN

the three regimes of the tree, in its colours: each frame carries its own, top right

UPPER	the formula, with any condition it needs
LOWER	what the construction achieves, as a formula too
PROVED	a bound with a proof behind it
COMPUTED	a value found by exhausting a finite search space, which is a computation and is never called a proof
1. 2. 3.	a sketch, upper bound first, ending with what attains it, then the results it leans on
EXTREMAL	the drawing is that witness, with the size and the shared value beside it, so it can be counted against that number

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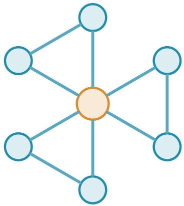
Simple graph · edge

$\ell_m(n)$

EXACT VALUE ESTABLISHED

UPPER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ PROVED $n \geq m$

LOWER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ PROVED



$n = 7, \ m = 3$ **9** EXTREMAL

1. a Gomory–Hu tree holds every pairwise connectivity in $n-1$ cuts
2. each has capacity at most $m-1$, so they carry $(m-1)(n-1)$ in all
3. every edge is charged to two of those cuts, which halves the count
4. a star with a degree-$(m-2)$ spoke graph on its leaves attains it

USES Mader 1.2 · Gomory–Hu A.2 · double count A.5–7 · hub-and-spokes A.8–9

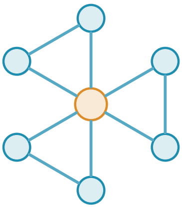
Simple graph · vertex

$k_m(n)$

PARTIAL OR CONJECTURAL

UPPER $k_m(n) = \ell_m(n)$ PROVED $m \le 4$

LOWER $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ PROVED



$n = 7,\ m = 3$ **9** EXTREMAL

1. the edge bound does not transfer: $\kappa \le \lambda$ makes vertex feasibility the weaker restriction, so there are more graphs to beat
2. for $m \le 4$ the two values are equal anyway, which is the cited result
3. at $m = 5$ they part: $\left\lfloor \frac{8n}{3} \right\rfloor - 4$ takes over from $n \ge 14$
4. the same drawing attains it, every leaf having degree two

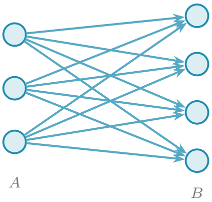
USES Bártfai–Bollobás–Leonard 1.4 · Sørensen–Thomassen 1.5 · degree bound A.3

Simple digraph · arc

$\ell_m^{\mathrm{dir}}(n)$

UPPER $M(n) = \max\left(2(n-1), \lfloor n^2/4 \rfloor\right)$ PROVED $m = 2$

LOWER $\left\lfloor \frac{n^2}{4} \right\rfloor$ PROVED



$n = 7,\ m = 2$ **12**

EXTREMAL

1. a reachability skeleton is a subdigraph preserving who reaches whom
2. at $m = 2$ a feasible digraph *is* its own skeleton: drop an arc and the route replacing it is a second disjoint one, breaching the cap
3. a skeleton carries at most $M(n)$ arcs, which bounds the digraph
4. at $n = 7$ both branches of M equal 12, so the bidirected hub and the one-way bipartite digraph each attain it

USES Arc value at $m = 2$ 1.6 · skeleton A.18–A.19 · two-hop A.20 · the two $m = 2$ families A.25 · directed hub A.15 · one-way bipartite A.16

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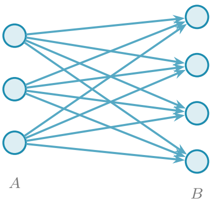
Simple digraph · vertex

$k_m^{\mathrm{dir}}(n)$

UPPER $M(n)$ PROVED $m=2$

LOWER $\left\lfloor \frac{n^2}{4} \right\rfloor$ PROVED

GENERAL VALUE OPEN



$n=7,\ m=2$ **12**

EXTREMAL

1. the routes the skeleton argument produces are internally vertex-disjoint, not merely arc-disjoint
2. so the argument proves the *vertex* bound first, and the arc bound is the consequence rather than the source
3. that direction matters: Whitney would not license carrying a bound the other way
4. the same digraph attains both, so the two values coincide at $m=2$

USES Vertex value at $m=2$ 1.7 · sparse skeleton A.19 · deletion monotonicity A.23 · one-way bipartite A.16

Multigraph · edge

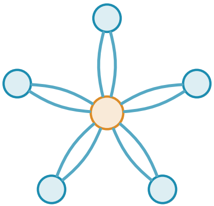
$L_m(n)$

EXACT VALUE ESTABLISHED

UPPER $(m-1)(n-1)$ PROVED every n and m

LOWER $(m-1)(n-1)$ PROVED

1. the same Gomory–Hu charge: $n-1$ cuts, each of capacity at most $m-1$
2. but parallel copies mean nothing forces an edge to be counted twice
3. so no halving and no floor: the bound is the clean product, and the missing factor of two is exactly what simplicity was buying next door
4. any spanning tree at multiplicity $m-1$ attains it



$n = 6,\ m = 3$ **10**

EXTREMAL

USES Multigraph edge value 1.3 · Gomory–Hu A.2 · thickened tree A.36

Multigraph · vertex

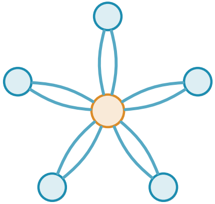
$K_m(n)$

PARTIAL OR CONJECTURAL

UPPER $2(n-1)$ PROVED $m = 3$

LOWER $(m-1)(n-1)$ PROVED

1. between two vertices κ is exactly $\mu(u,v) + \pi(u,v)$: the copies of their own edge, plus the disjoint detours around it
2. feasibility therefore caps those two *together* at $m-1$
3. at $m = 3$ that yields the same value as the edge problem
4. a tree edge has no detour, so a doubled tree spends the whole cap on multiplicity and attains it



$n = 6,\ m = 3$ **10**

EXTREMAL

USES Splitting the vertex measure A.54 · value at $m = 2$ A.55 · value at $m = 3$ A.57 · incidence rank A.52

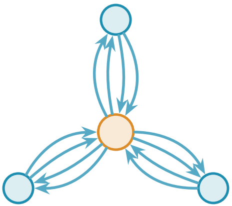
Directed multigraph · arc

$L_m^{\mathrm{dir}}(n)$

EXACT VALUE ESTABLISHED

UPPER $(m-1)M(n)$ PROVED every n and m

LOWER $(m-1)\cdot 2(n-1)$ PROVED



$n=4,\ m=3$ **12**

EXTREMAL

1. inside any feasible multigraph there is a sparse reachability skeleton
2. removing one copy of each of its arcs costs every reachable pair one route, so the graph stays feasible one threshold lower
3. $m-1$ such peels account for every arc, giving $m-1$ times the $m=2$ value, and the fractional relaxation is integral so nothing is lost
4. a bidirected tree at multiplicity $m-1$ attains the linear branch

USES Every n and m 1.13 · sparse skeleton A.19 · deleting a skeleton A.37 · thickened tree A.36 · integrality A.39

Directed multigraph · vertex

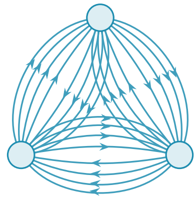
$K_m^{\mathrm{dir}}(n)$

GENERAL VALUE OPEN

UPPER $(m-1)\left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1)$ PROVED reads 210 here, far from tight

LOWER $n(n-1)(m+1-n)$ PROVED

1. κ splits as $\mu+\pi$ again: direct copies plus disjoint detours
2. where all six directions are present each pair has exactly one detour, so its multiplicity stops at $m-2=4$ rather than at the cap of five
3. the six directions give $6\times4=24$, and enumeration leaves exactly one isomorphism class
4. it is the complete digraph thickened, the $b=3$ block of A.65



$n=3,\ m=6$ **24**

COMPUTED EXTREMAL

USES Directed incidence value A.64 · splitting the vertex measure A.54 · thickened complete digraph A.65

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Simple hypergraph · edge

$\ell_m^{(r)}(n)$

PARTIAL OR CONJECTURAL

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ PROVED

LOWER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ PROVED $m-1 \leq \binom{n-2}{r-2}$

$n=9,\ r=3,\ m=2$ **4**

EXTREMAL

1. induction on a single hyperedge cut, with no tree anywhere in it
2. a hyperedge spans r vertices but still serves only one route, so cutting one off divides what remains by $r-1$
3. the reduction to graphs points the wrong way: replacing a hyperedge by a star would let two routes take two of its edges, which Berge forbids
4. the star hypertree attains it, and for $r \geq 3$ simplicity is free

USES Hypergraph edge bound 1.8 · cut induction A.45 · simplicity is free A.46 · sparse hypergraphs A.49 · lower bound A.50

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Simple hypergraph · vertex

$k^{(r)}_m(n)$

PARTIAL OR CONJECTURAL

UPPER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ PROVED $m = 2$

LOWER $\left\lfloor \frac{n-1}{r-1} \right\rfloor$ PROVED

$n = 9,\ r = 3,\ m = 2$ **4**

EXTREMAL

1. at $m = 2$ feasibility says exactly that the incidence graph is a forest, since a cycle in it would already be two disjoint routes
2. a forest on the n vertices together with the $|E|$ hyperedges has at most $n + |E| - 1$ edges
3. each hyperedge contributes r of them, so $r|E| \leq n + |E| - 1$, which rearranges to $|E|(r-1) \leq n - 1$
4. the star hypertree's incidence graph is a tree, so it attains it

USES Vertex value at $m = 2$ 1.9 · at $m = 3$ 1.10 · incidence rank A.52 · Menger for multihypergraphs A.44

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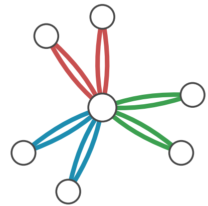
Multihypergraph · edge

$L_m^{(r)}(n)$

PARTIAL OR CONJECTURAL



$n = 7,\ r = 3,\ m = 3$ **6**

EXTREMAL

UPPER $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ PROVED unchanged by repetition

LOWER $\frac{(m-1)(n-1)}{r-1}$ PROVED $(r-1) \mid (n-1)$

1. the cut induction never asks whether hyperedges are distinct
2. so the bound is word for word the one next door, and repetition cannot move it
3. what repetition changes is which shapes reach it: the divisibility condition replaces the supply of distinct hyperedges
4. the star hypertree with each hyperedge taken $m-1$ times attains it

USES Hypergraph edge bound 1.8 · cut induction A.45 · multihypergraph lower bound A.51 · twelve copies on six vertices A.48

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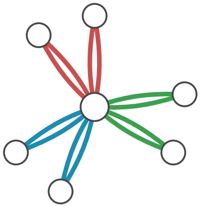
Multihypergraph · vertex

$K_m^{(r)}(n)$

PARTIAL OR CONJECTURAL

UPPER $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ PROVED $m=3$

LOWER $\frac{(m-1)(n-1)}{r-1}$ PROVED $(r-1) \mid (n-1)$



$n=7,\ r=3,\ m=3$ **6**

EXTREMAL

1. the incidence graph again, but now its cycle rank is what needs bounding
2. the rank is controlled by decomposing the graph at its cut vertices and at its separating pairs
3. repetition buys nothing at $m=2$, where two copies of one hyperedge are already two separated routes
4. from $m=3$ it adds attainment cases, and the repeated star is one

USES Vertex value at $m=3$ 1.10 · incidence rank A.52 · multihypergraph lower bound A.51

Directed simple hypergraph · arc

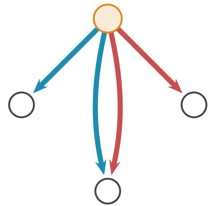
$\ell_m^{\mathrm{dir}\,(r)}(n)$

UPPER $\dfrac{(m-1)n(n-1)}{r-1}$ PROVED reads 12 here, not tight

LOWER $n \cdot \min\left(m-1, \binom{n-1}{r-1}\right)$ PROVED

1. every route out of a vertex has to start with one of that vertex's own outgoing hyperedges
2. deleting those cuts it off, so the outgoing count alone caps every pair it reaches, and giving each vertex $\min(m-1, \binom{n-1}{r-1})$ of them is feasible at any size
3. for general n the matching upper bound is only asymptotic: a shadow digraph carries the route cap onto two-step routes
4. at this size enumeration closes the gap, and $4 \times 2 = 8$

GENERAL VALUE OPEN



$n = 4, \ r = 3, \ m = 3$ **8**

COMPUTED EXTREMAL

one tail of the four, each doing the same

USES Directed hypergraph constant 1.14 · first bounds A.67 · bounded outgoing count A.68 · one-step shadow A.71 · two-step budget A.28

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Directed simple hypergraph · vertex

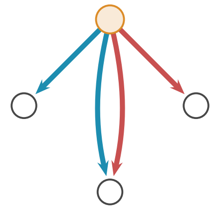
$k_m^{\mathrm{dir}\,(r)}(n)$

UPPER $\dfrac{(m-1)n(n-1)}{r-1}$ **PROVED** reads 12 here, not tight

LOWER $n \cdot \min\left(m-1, \binom{n-1}{r-1}\right)$ **PROVED**

1. the outgoing cut never looks at interior vertices at all
2. so the very same argument establishes the arc bound and the vertex bound in one go, and the witness needs no adjustment
3. the shadow argument behind the general growth rate re-runs under the stricter separation for the same reason
4. whether the two separations keep one value at larger sizes is open

GENERAL VALUE OPEN



$n = 4,\ r = 3,\ m = 3$ **8**

COMPUTED **EXTREMAL**

one tail of the four, each doing the same

USES Directed hypergraph constant 1.14 · first bounds A.67 · bounded outgoing count A.68

KU LEUVEN

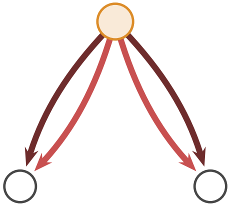
Directed multihypergraph · arc

$L_m^{\mathrm{dir}\,(r)}(n)$

GENERAL VALUE OPEN

UPPER $\dfrac{(m-1)n(n-1)}{r-1}$ PROVED

LOWER $n(m-1)$ PROVED



$n=3,\ r=3,\ m=3$ **6**

EXTREMAL

one tail of the three, each doing the same

1. counting tail–head pairs gives the upper bound: each hyperedge covers $r-1$ of them, and no ordered pair takes more than $m-1$
2. the outgoing cut caps each tail at $m-1$ copies, whatever those copies are, so with repeats allowed the construction reaches $n(m-1)$
3. at this size a tail has only one possible head set, the other two vertices, so repetition is the only way to grow at all
4. the two meet here, both reading 6, so the value needs no search: it is the smallest size at which the counting bound is already tight

USES Directed hypergraph constant 1.14 · first bounds A.67 · bounded outgoing count A.68 · backward reverses forward A.69

KU LEUVEN

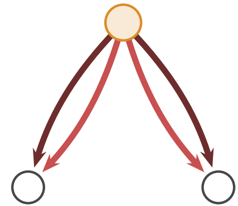
Directed multihypergraph · vertex

$K_m^{\mathrm{dir}\,(r)}(n)$

UPPER $\dfrac{(m-1)n(n-1)}{r-1}$ PROVED

LOWER $n(m-1)$ PROVED

GENERAL VALUE OPEN



$n=3,\ r=3,\ m=3$ **6**

EXTREMAL

one tail of the three, each doing the same

1. the pair count bounds this variant too, and the outgoing cut ignores interior vertices, so both bounds carry over from the arc problem unchanged
2. they meet here as well, so this size is settled by argument rather than by search
3. the general model, which allows every non-empty tail–head split, obeys the forward bound and shares its constant
4. whether forward and general agree at a fixed $n > r$, and not merely in the growth rate, is open

USES General orientation constant A.72 · general obeys forward A.70 · backward reverses forward A.69 · bounded outgoing count A.68